Miderm Exam I (Take Home)

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6.

(a) The fact that S is closed under $\star$ comes from the fact that (with notation as in the problem statement) g_1g_2 and $g_1g_2^{-1}$ are both in G by the closure of G, and $h_1h_2 \in H$ by the closure of H, and so $(g_1g_2^{h_1}, h_1h_2) \in G \times H = S$. Associativity comes by the following argument:

$$\begin{aligned}((g_1, h_1) \star (g_2, h_2)) \star (g_3, h_3) &= (g_1g_2^{h_1}, h_1h_2) \star (g_3, h_3)\\ &= (g_1g_2^{h_1}g_3^{h_1h_2}, h_1h_2h_3)\\ &= (g_1g_2^{h_1}g_3^{h_2h_1}, h_1h_2h_3)\\ &= (g_1(g_2g_3^{h_2})^{h_1}, h_1h_2h_3)\\ &= (g_1, h_1) \star (g_2g_3^{h_2}, h_2h_3)\\ &= (g_1, h_1) \star ((g_2, h_2) \star (g_3, h_3))\end{aligned}$$

The identity is $(e, 1)$, where e is the identity in G, since given $(g, h) \in S$,

$$(g, h) \star (e, 1) = (g \cdot e^h, h \cdot 1) = (g, h).$$

Finally, the inverse of (g, h) is (g^{-h}, h^{-1}) since

$$\begin{aligned}(g, h) \star (g^{-h}, h^{-1}) &= (g(g^{-h})^h, hh^{-1})\\ &= (gg^{-h^2}, 1)\\ &= (gg^{-1}, 1)\\ &= (e, 1).\end{aligned}$$

(b) Take $(g, 1), (h, 1) \in \widehat{G}$. Then using the inverse formula from part (a), we show

$$(g, 1) \star (h^{-1}, 1) = (gh^{-1}, 1) \in S$$

since $gh^{-1} \in G$. Thus $\widehat{G}$ is a subgroup. To show that it is normal, consider $(g, 1) \in \widehat{G}$

and $(a,b) \in S$. Then

$$\begin{aligned}(a,b) \star (g,1) \star (a,b)^{-1} &= (a,b) \star (g,1) \star (a^{-b}, b^{-1}) \\ &= (ag^b, b) \star (a^{-b}, b^{-1}) \\ &= (ag^b a^{-b^2}, 1) \\ &= (ag^b a^{-1}, 1) \\ &= (aa^{-1} g^b, 1) \\ &= (g^b, 1) \in \widehat{G}.\end{aligned}$$

(c) Using the last line of part (b), we see that conjugacy classes of elements of $\widehat{G}$ consist of two elements each since each is of the form $\{(g,1),(g^{-1},1)\}$.

8.

(a) Using parts (b) and (c), we see that $|Z(G)| = 2$ and $|G/Z(G)| = 8$, so by Lagrange's theorem, $|G| = 16$.

(b) We want to find a general element $x^a y^b$ such that $(x^a y^b)(x^c y^d) = (x^c y^d)(x^a y^b)$ for any appropriate choices of c and d (i.e. c is between 0 and 7 and d is either 0 or 1). If our b is 0, then the equation becomes

$$x^{a+c} y = x^c y x^a = x^{c+3a} y,$$

which after multiplying on the right by y^{-1}, x^{-c}, and x^{-a}, gives us $1 = x^{2a}$. This is true whenever $a = 0$ or $a = 4$, so 1 and x^4 are in $Z(G)$.

When $b = 1$, we can do a similar calculation when $d = 1$ to get

$$\begin{aligned} & x^a y x^c y = x^c y x^a y \\ \iff & x^{a+3c} y^2 = x^{c+3a} y^2 \\ \iff & x^{a+3c} = x^{c+3a} \\ \iff & x^{2c} = x^{2a}.\end{aligned}$$

Since this is dependent on the value of c, we conclude that when $b = 1$, $x^a y$ does not commute with every element, and thus $Z(G) = \{1, x^4\}$.

(c) This gives us a natural surjective homomorphism from G to $\{x, y : x^4 = y^2 = 1, yx = x^3 y\}$ where $Z(G)$ is the kernel, so we can identify $G/Z(G)$ with D_8.

9.

(a) Indeed, the identity homomorphism id_G acts as the identity since $\mathrm{id}_G \cdot g = \mathrm{id}_G(g) = g$, and given $\varphi, \psi \in \mathrm{Aut}(G)$,

$$\begin{aligned}\varphi \cdot (\psi \cdot g) &= \varphi \cdot \psi(g) \\ &= \varphi(\psi(g)) \\ &= (\varphi \circ \psi)(g) \\ &= (\varphi \circ \psi) \cdot g,\end{aligned}$$

so this is an action.

(b) If $H = \bigcup_{g \in A \subset G} \operatorname{Orb}(g)$, and given an automorphism $\varphi \in \operatorname{Aut}(G)$ and $h \in H$, then $h \in \operatorname{Orb}(g_h)$ for some $g_h \in A$. Since orbits partition G and $\operatorname{Orb}(h) \cap \operatorname{Orb}(g_h) \neq \varnothing$, we must have $\operatorname{Orb}(h) = \operatorname{Orb}(g_h)$. Thus

$$\varphi \cdot h = \varphi(h) \in \operatorname{Orb}(h) = \operatorname{Orb}(g_h) \subset H,$$

so $\varphi \cdot H = H$.

On the other hand, if H is characteristic in G, then we want to show that for any $h \in H$, $\operatorname{Orb}(h) \subset H$. We can represent a general element of $\operatorname{Orb}(h)$ as $\varphi(h)$ for some $\varphi \in \operatorname{Aut}(G)$. Since H is characteristic, $\varphi(H) = H$, so in particular $\varphi(h) \in H$, and we are done.

(c) When G is finite cyclic of order n, presented multiplicatively as $\langle x : x^n = 1 \rangle$, the automorphisms are exactly those homomorphisms which send x to x^m, where m is relatively prime to n. Thus the orbit of x is $\{x^m : \gcd(m, n) = 1,\, 0 < m < n\}$. The identity will always be sent to the identity, so another orbit is just $\{1\}$. The other orbits will be orbits of x^a where $2 \leq a < n$, and these orbits can be represented as $\{x^{am} : \gcd(m, n) = 1,\, 0 < m < n\}$.

10.

(a) Given two morphisms $\varphi : X \to Y$ and $\psi : Y \to Z$, we define $\psi \circ \varphi$ as function composition, namely $(\psi \circ \varphi)(x) = \psi(\varphi(x))$. This is indeed a morphism since given $g \in G$ and $x \in X$,

$$\begin{aligned}(\psi \circ \varphi)(g \cdot x) &= \psi(\varphi(g \cdot x)) \\ &= \psi(g \cdot \varphi(x)) \\ &= g \cdot \psi(\varphi(x)) \\ &= g \cdot (\psi \circ \varphi)(x).\end{aligned}$$

The identities id_X and id_Y act as identities with respect to this composition since

$$\mathrm{id}_Y(\varphi(x)) = \varphi(x)$$

and

$$\varphi(\mathrm{id}_X(x)) = \varphi(x),$$

and these identites are in fact morphisms since they are G-equivariant.

(b) Let $a \in \operatorname{Orb}_G(\varphi(x))$. Then there exists a $g \in G$ such that

$$a = g \cdot \varphi(x) = \varphi(g \cdot x) \in \varphi(\operatorname{Orb}_G(x)).$$

Similarly, given $b \in \varphi(\operatorname{Orb}_G(x))$, there exists a $g \in G$ such that

$$b = \varphi(g \cdot x) = g \cdot \varphi(x) \in \operatorname{Orb}_G(\varphi(x)).$$

Thus $\operatorname{Orb}_G(\varphi(x)) = \varphi(\operatorname{Orb}_G(x))$.

(c) Let $g \in \operatorname{Stab}_G(x)$. Then $g \cdot x = x$, so $g \cdot \varphi(x) = \varphi(g \cdot x) = \varphi(x)$, so $g \in \operatorname{Stab}_G(\varphi(x))$. Now let $h \in \operatorname{Stab}_G(\varphi(x))$. Then $\varphi(x) = h \cdot \varphi(x) = \varphi(h \cdot x)$. If φ is not injective, we may not be able to go any further, but if it is, then we can left-cancel φ and get $x = h \cdot x$, and thus $h \in \operatorname{Stab}_G(x)$.